Adam Schroeder

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Education

PhD Mechanical and Aerospace Engineering, UCLA Fall 2025-present

Major: Fluid Mechanics

Minor: Dynamical Systems and Control

GPA: 4.0/4.0

B.A. Mathematics, Macalester College

Minor: Computer Science May 2025

GPA: 4.0/4.0

Leadership: Resident Advisor 2022-2023

Membership: NCAA Division III Swim 2021-2025

Relevant Coursework:

Honors Thesis in Turbulence Modeling, Independent Study in Finite Volumes Methods, Fluid Mechanics for CE, Mathematical Modeling, Mathematical Statistics, Computational Linear Algebra, Topology, Algorithms, Data Structures

Research Experience

Nonlinear and nonmodal stability analysis of time-varying flows, *Fall 2025-present*

UCLA

- Generalize nonmodal, nonlinear methods to dynamic base states
- Identify codimension-one invariant manifolds between laminar and turbulent basins of attraction

Stability of turbulent boundary layers, *Fall 2024-Spring 2025*

Macalester College

- Independently developed a mathematical background in turbulence theory and PDE-theoretical dynamics
- Modeled benchmark turbulent flat plate flows using RANS k-ω SST models and validated against NASA's 2D zero-pressure flat plate verification data
- Implemented the Arnoldi iterations method to recover the leading unstable modes from the base flow

Simulations of contrail lifecycles, *May 2024-August 2024*

Deutsches Zentrum für Luft- und Raumfahrt (DLR), Oberpfaffenhofen

- Modeled early contrail development in descending wake vortices using the non-hydrostatic, anelastic LES EULAG solver coupled with Lagrangian particle tracking for ice microphysics (LCM), using both analytic initial profiles as well as profiles resulting from RANS simulations
- Analyzed the sensitivity of hydrogen combustion contrail physics to variations in atmospheric and vehicle parameters, including scaling the initial ice crystal number and varying the number of plumes and their respective radii

- Devised and implemented an optimized circulation routine which provides tighter margins of error compared to previous methods

Research assistant in topological data analysis, *May 2023-July 2024*
Carlson School of Management (University of Minnesota) and Macalester College

- Adapted the techniques used in topological data analysis to knowledge networks from the science of science and presented the preliminary results at the MAA-NCS conference (Mathematical Association of America)
- Devised and implemented linear programs for thresholding bibliometric data by exploiting persistent homology, including a constrained minimization-maximization problem that attempts to minimize distance between persistence diagrams as measured by Euclidean norms between vectorized persistence images constrained by a minimum number of homology features
- Wrote a manuscript detailing the applications of TDA to knowledge networks

Teaching Experience

Teaching Assistant, *Spring 2025*
Mathematical Statistics, Macalester College

- Bridged the knowledge gap between applied statistics and mathematical formalism through review sessions and developing new practice and homework problems in a short textbook still used by the instructor

Teaching Assistant, *Fall 2024*
Partial Differential Equations, Macalester College

- Facilitated the reintroduction of the course to the curriculum and aided students in learning the theory and application of paradigmatic PDEs

Teaching Assistant, *Spring 2024*
Introduction to Database Management Systems, Macalester College

- Reinforced students with fundamental concepts underlying relational databases systems, covering LDS models and relational operators, and facilitated students' database and query constructions in MariaDB
- Held regular study sessions throughout the week and participated in monthly workshops to brainstorm peer-learning strategies

Teaching Assistant, *Fall 2023*
Discrete Mathematics, Macalester College

- Developed students' comfortability with fundamental concepts in proof-based mathematics

Conferences and Manuscripts

Schroeder, A., Guan, J. (September 2023). *Topological Data Analysis of Knowledge Networks*. MAA-NCS Fall 2023 Section Meeting, University of Minnesota Duluth.

Schroeder, A., Funk, R., Guan, J., Owen-Smith, J., Ziegelmeier, L. (January 2024). *Topological Data Analysis of Knowledge Networks*. Joint Mathematics Meetings, San Francisco.

Schroeder, A., Guan, J., Funk, R., Ziegelmeier, L. (March 2025). *Higher-Order Network Structure Inference: A Topological Approach to Threshold Selection*. MAA-NCS Spring 2025 Section Meeting, St. Olaf College.

(Poster) **Adam Schroeder**, Lori Ziegelmeier, Russell Funk, Jingyi Guan. *Higher-Order Structure Inference: A Topological Approach to Parameter Selection*. ATMCS 11, Bozeman MT.

Adam D. Schroeder and Lori Ziegelmeier, Macalester College, U.S.; Jingyi Guan, University of Washington, U.S.; Russell Funk, University of Minnesota, U.S., *Higher-Order Network Structure Inference: A Topological Approach to Threshold Selection*. SIAM AN25, Montreal.

(Preprint) **Adam Schroeder**, Russell Funk, Jingyi Guan, Taylor Okonek, & Lori Ziegelmeier. (2025). *Higher-Order Network Structure Inference: A Topological Approach to Network Selection*, arXiv:2510.04884

(In-Progress) **Schroeder, A.**, Funk, R., Gebhart, T. *Expectation of the CD Index*.

Awards and Honors

Awards
National Science Foundation (NSF) Graduate Research Fellowships Program (GRFP) Honorable Mention, Konhauser Achievement Award in Mathematics (Macalester College), honors student in mathematics, three-time NCAA DIII Academic All-District team, three-time Academic All-MIAC team, Lowell Thomas public speaking prize (Macalester College), Dean's list (Macalester College)
Scholarships
2024 Deutscher Akademischer Austauschdienst (DAAD) RISE scholarship for science and engineering, DeWitt Wallace scholarship (Macalester College), Kay Thomas Scholar (Macalester College)

Memberships

Society of Industrial and Applied Mathematics (2024-25)

Skills

Languages: English (native), Dutch (fluent), German (fair)
Programming: Python, C++, R, Java, MariaDB, PostgreSQL, Fortran, Mathematica, OpenFOAM. MSI, DoD Carpenter, Hoffman2, DKRZ supercomputing experience (Linux OS)